

AI SERVICE-CALL SIMULATOR • HVAC

FIELDREADY

Practice the service call before it is real.

UNIT Carrier 24ACC636

SCENARIOS 8

CHECKS 22

READINGS Simulated

THE PROBLEM

THE PRACTICE GAP

Diagnostic judgment comes from repetition, and the calls that teach the most are the hardest to stage.

Rare faults

A trainee cannot count on meeting an open fan-motor winding during training.

Unsafe to reproduce

Creating a real fault on energized equipment puts learners at risk.

Limited lab time

Hands-on sessions are short, and there is little to practice on between them.

HOW IT WORKS

A PRACTICE CALL IN FIVE STEPS

01

Receive

A service request arrives, such as an air conditioner blowing warm air.

02

Interview

The trainee questions an AI homeowner about symptoms and history.

03

Check

They state an inspection or measurement and get a reading that fits the hidden fault.

04

Diagnose

They name the cause, cite the evidence and recommend a repair.

05

Debrief

A grading agent scores safety, order, evidence and the fix.

The homeowner never names the fault, and the simulator never coaches. The trainee has to reason their way there.

ONE CALL, START TO FINISH

SERVICE REQUEST

Air conditioner cools for a few minutes, then blows warm. Outside unit sounds different than usual.

Carrier 24ACC636, 3-ton split system

88°F outside, 79°F inside

Hidden fault: outdoor fan motor has an open winding

HOMEOWNER	It starts out cool and then goes warm. And the outside unit is loud but I don't feel any air coming off the top like I usually do.
TRAINEE	I'll look and listen at the outdoor unit from a safe distance.
READING	Compressor is running but the fan blade is not turning. After a few minutes the compressor shuts off.
TRAINEE	Disconnect off, locked and tagged, zero volts verified. Checking the fan motor windings.
READING	Open circuit between two of the three leads, even after the motor has cooled.

EIGHT CALLS ON ONE UNIT

What the homeowner reports	Hidden fault	Level
Running but blowing warm air	Compressor side of the dual run capacitor failed	Beginner
Outside unit does nothing	Contactor coil open	Beginner
Stopped after work outside	Outdoor disconnect left OFF	Beginner
Runs constantly, weak airflow	Clogged filter, frosted indoor coil	Beginner
Cools briefly, then warm	Outdoor fan motor open winding	Intermediate
Warm on hot afternoons only	Outdoor coil blocked with debris	Intermediate
Outside unit does nothing	Low-voltage cable cut	Intermediate
Runs nonstop, only slightly cool	Undercharged from a slow leak	Advanced

All eight share the same 22 checks, so the menu never gives the answer away. Only the results differ.

WHERE THE NUMBERS COME FROM

READINGS GROUNDED IN THE MANUALS

4

Carrier documents: service manual, installation instructions, wiring diagrams, product data

CITED • Service manual p. 11

26 V across the contactor coil. Normal is 20 to 30 V.

2

OSHA publications for safety framing: electrical hazards and lockout/tagout

22

Checks per scenario, each tagged with the manual page or marked as an assumption

ASSUMPTION • Listed in the brief

Dual run capacitor is rated 45/5 μ F, 440 V. The manuals do not state the rating.

A SCORECARD IN FOUR AREAS



Safety

Locks out, verifies zero voltage and discharges the capacitor before risky steps.



Order

Moves from simple, likely checks to specific ones without guessing or swapping parts.



Evidence

Cites the key readings and rules out at least one other plausible cause.



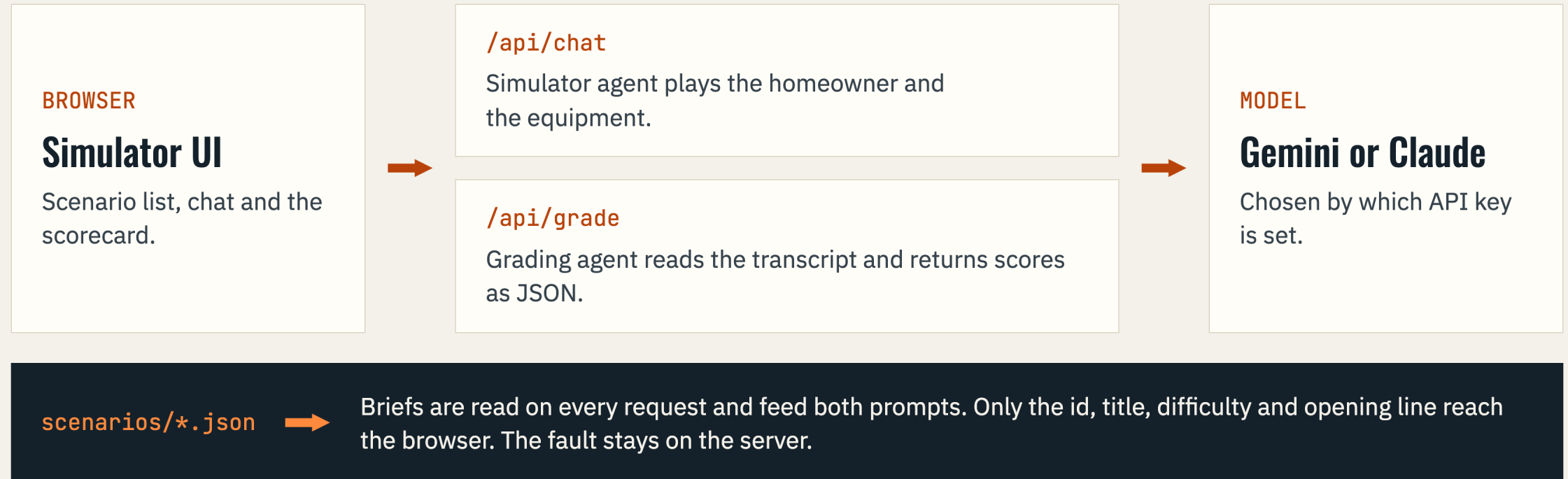
Fix

Names the actual fault and a complete, correct repair.

Each area is scored 0 to 100 with a one-sentence note, and the overall score is their average. A correct guess with no supporting readings scores low on evidence.

ARCHITECTURE

A NEXT.JS APP WITH TWO AGENTS



STATUS

KNOWN GAPS AND NEXT STEPS

Open today

The grader has not been run end to end against the eight scenarios.

Every attempt at a scenario is identical, with no variation yet.

No HVAC professional has reviewed the briefs.

Checks are typed as free text, so there is no structured log of what was run.

Next

Tune the grader prompt on real transcripts.

Vary the homeowner's account and the readings while the fault stays the same.

Feed the grader the citations and safety prerequisites already in each brief.

Add voice. The mic button is a placeholder today.

FIELDREADY

PRACTICE THE SERVICE CALL BEFORE IT IS REAL.

`npm install` `npm run dev` `localhost:3000`